

Task No.: 7**Date: 08-02-2026**

Title: Develop a native application that uses GPS location information**Aim:**

To develop an Android Application that uses GPS location information.

Code:*MainActivity.kt*

```
package com.example.madtask8

import android.Manifest
import android.content.pm.PackageManager
import android.location.Geocoder
import android.os.Bundle
import android.widget.Button
import android.widget.TextView
import androidx.activity.result.contract.ActivityResultContracts
import androidx.appcompat.app.AppCompatActivity
import androidx.core.app.ActivityCompat
import com.google.android.gms.location.*
import java.util.Locale

class MainActivity : AppCompatActivity() {

    private lateinit var txtLocation: TextView
    private lateinit var txtAddress: TextView
    private lateinit var btnGetLocation: Button

    private lateinit var fusedLocationClient: FusedLocationProviderClient
    private lateinit var locationCallback: LocationCallback

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        txtLocation = findViewById(R.id.txtLocation)
```

```

txtAddress = findViewById(R.id.txtAddress)
btnGetLocation = findViewById(R.id.btnGetLocation)

fusedLocationClient = LocationServices.getFusedLocationProviderClient(this)

btnGetLocation.setOnClickListener {
    checkPermissionAndGetLocation()
}

locationCallback = object : LocationCallback() {

    override fun onLocationResult(result: LocationResult) {

        val location = result.lastLocation

        if (location != null) {

            val latitude = location.latitude
            val longitude = location.longitude

            txtLocation.text =
                "Latitude: $latitude\nLongitude: $longitude"

            val geocoder = Geocoder(this@MainActivity, Locale.getDefault())

            val addressList =
                geocoder.getFromLocation(latitude, longitude, 1)

            if (!addressList.isNullOrEmpty()) {

                val address = addressList[0]

                txtAddress.text =
                    "${address.locality}, ${address.adminArea}, ${address.countryName}"

            } else {

                txtAddress.text = "Address not found"

            }

        }
    }
}

```

```

        fusedLocationClient.removeLocationUpdates(locationCallback)
    }
}
}
}

```

```

private fun checkPermissionAndGetLocation() {

    if (ActivityCompat.checkSelfPermission(
        this,
        Manifest.permission.ACCESS_FINE_LOCATION
    ) != PackageManager.PERMISSION_GRANTED
    ) {

        requestPermissionLauncher.launch(
            Manifest.permission.ACCESS_FINE_LOCATION
        )

    } else {

        getLocation()

    }
}

```

```

private fun getLocation() {

    val locationRequest = LocationRequest.Builder(
        Priority.PRIORITY_HIGH_ACCURACY,
        1000
    )
        .setMaxUpdates(1)
        .build()

    if (ActivityCompat.checkSelfPermission(
        this,
        Manifest.permission.ACCESS_FINE_LOCATION
    ) != PackageManager.PERMISSION_GRANTED
    ) return

    fusedLocationClient.requestLocationUpdates(

```

```

        locationRequest,
        locationCallback,
        mainLooper
    )
}

private val requestPermissionLauncher =
    registerForActivityResult(
        ActivityResultContracts.RequestPermission()
    ) { granted ->

        if (granted) {

            getLocation()

        } else {

            txtAddress.text = "Permission Denied"

        }
    }
}

```

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp">

    <!-- Top Section (Name and Roll No) -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:gravity="center_horizontal"
        android:orientation="vertical">

        <TextView

```

```

    android:id="@+id/txtName"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Name: Yoga Shri P K"
    android:textSize="20sp"
    android:textStyle="bold"
    android:layout_marginTop="10dp"/>

```

```

<TextView
    android:id="@+id/txtRollNo"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Roll No.: 23BCS192"
    android:textSize="18sp"
    android:textStyle="bold"/>

```

```

</LinearLayout>

```

```

<!-- Center Section (Location info centered vertically and horizontally) -->

```

```

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="0dp"
    android:layout_weight="1"
    android:gravity="center"
    android:orientation="vertical">

```

```

<TextView
    android:id="@+id/txtLocation"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Latitude: -- \nLongitude: --"
    android:textSize="18sp"
    android:textStyle="bold"
    android:gravity="center"
    android:layout_marginBottom="10dp"/>

```

```

<TextView
    android:id="@+id/txtAddress"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Address: --"
    android:textSize="16sp"

```

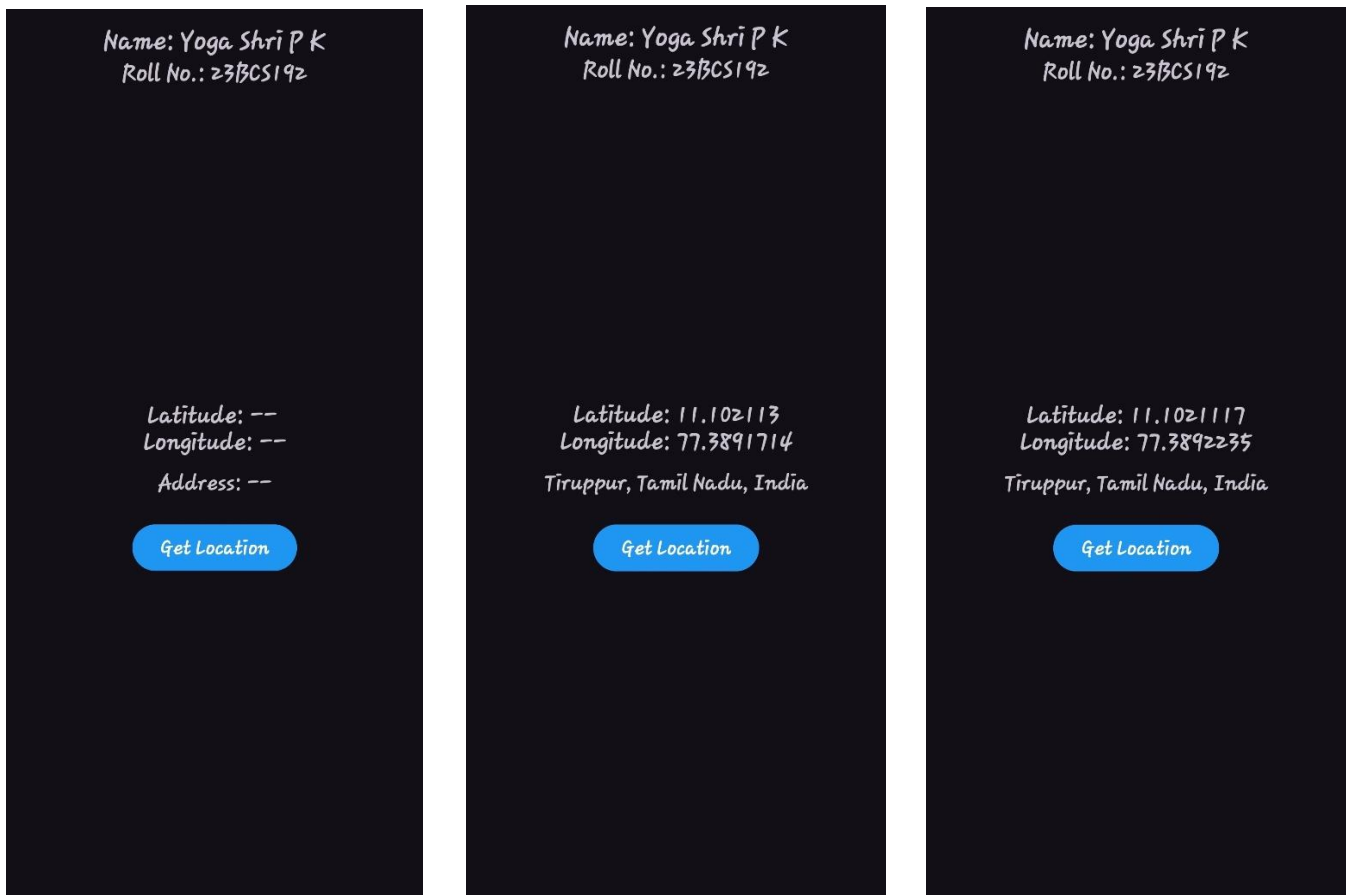
```
android:gravity="center"  
android:layout_marginBottom="20dp"/>
```

```
<Button  
    android:id="@+id/btnGetLocation"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="Get Location"  
    android:textColor="@android:color/white"  
    android:backgroundTint="#2196F3"/>
```

```
</LinearLayout>
```

```
</LinearLayout>
```

Output:



Result:

Thus, Android Application that implements GPS location is developed and executed successfully.